★★ Not writing things I already know to shorten info ★★

1.1 → The fundamental insurance equation

→ Relationship between WP + EP
- → WP > EP ⇒ high growth
- → WP < EP ⇒ shrinking

→ Incurred (reported) loss = Paid loss + loss reserve

→ Estimated ultimate loss = Incurred loss + IBNR reserve + IBNER reserve

→ Underwriting profit = Income − Expenses (from strictly business on policies)

→ Price = Cost + Profit

Price ↓ premium; Cost ↓ {losses, LAE, UW expenses}; Profit ↓ UW profit

→ When balancing the fundamental insurance equation, need to address two considerations:
- → 1) Ratemaking is prospective → Goal is to estimate future, unknown costs w/ historical data, factoring in expected costs during the period the premium will be in effect
- → 2) Balance both aggregate + individual levels
  - ↳ premium should reflect the risk of the specific policyholder
  - ↳ total premium should be enough to cover all expected costs from entire book

⇒ If successful, then rates are adequate & equitable

→ Pure Premium (or loss costs) $= \frac{\text{Losses}}{\text{Exposures}} =$ Frequency * Severity

→ Average Premium $= \frac{\text{Premium}}{\#\text{ exposures}}$ (★ both earned or both written)

→ Loss ratio $= \frac{\text{Losses}}{\text{Premium}} = \frac{\text{Pure Premium}}{\text{Avg Premium}}$

→ Loss & LAE ratio $= \frac{\text{Loss + LAE}}{\text{Premium}}$

→ LAE ratio $= \frac{\text{LAE}}{\text{Losses}}$

→ Loss Ratio (1 + LAE ratio) $= \frac{\text{Losses}}{\text{Premium}}\left(1 + \frac{\text{LAE}}{\text{Losses}}\right)$
$= \downarrow \left(\frac{\text{Losses + LAE}}{\text{Losses}}\right)$
$= \frac{\text{Losses + LAE}}{\text{Premium}}$
$=$ Loss & LAE ratio

→ UW expense ratio $= \frac{\text{UW Expenses}}{\text{Premium}}$
- Incurred at inception ① → All except ② ⇒ ÷ by WP
- ------ during policy term ② → General expenses ÷ by EP

→ Operating expense ratio (OER) = UW expense ratio + $\frac{\text{LAE}}{\text{EP}}$
- → measures portion of each premium dollar used towards paying claim related expenses & UW expenses

→ Combined ratio $= \frac{\text{Losses}}{\text{EP}} + \frac{\text{LAE}}{\text{EP}} + \frac{\text{UW expenses}}{\text{WP or EP}}$
$=$ Loss + LAE ratio + ↓
$=$ Loss ratio + OER

1.2 → Policies, coverages + claims

→ Criteria for exposure bases
- → 1) proportional to expected loss → the factor that is most directly proportionate to losses (ie 10% increase in exposure ⇒ 10% increase in losses)
- → 2) practical → objective, easy/inexpensive to obtain + verify
- → 3) historical precedence → Should consider any preexisting exposure base established in the industry. Changes need to be carefully considered

→ Common exposure bases → The table below shows some commonly used exposure bases for different lines of business.

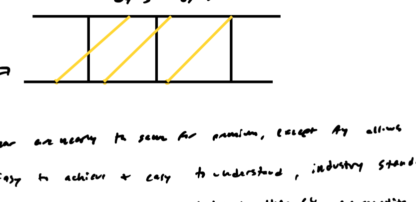
Line of Business	Typical Exposure Bases
Personal Automobile	Earned Car Year
Homeowners	Earned House Year
Workers Compensation	Payroll
Commercial General Liability	Sales Revenue, Payroll, Square Footage, Number of Units
Commercial Property	Amount of Insurance Coverage
Professional Liability	Number of Professionals
Personal Articles Floater	Value of Item

1.3 → Understanding insurance data

→ Data aggregation goals
- → 1) Accurately match losses + premium
- → 2) Use the most recent data available
- → 3) Minimize the cost associated w/ gathering & retrieving data

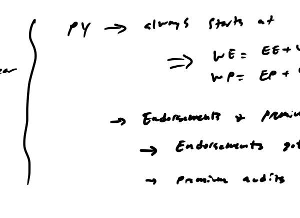
→ Four methods of data aggregation:

→ Calendar year aggregation
- → Advantages → data is readily available once the CY ends
  - → There is no future development (fixed at the end of the CY)
  - → Data is easily accessible
- → Disadvantages → mismatch between premium + losses (EP + losses can come from policies written in prior year)
  - → Inability to capture major developments due to the fixed nature of data



→ Accident year aggregation
- → Calendar / Accident year are nearly the same for premium, except AY allows for premium audits after the end of the CY
- → Advantages → Easy to achieve + easy to understand, industry standard
  - → Better match of premiums + losses than CY aggregation (ie losses year ** are compared to premium earned year **)
  - → Useful for identifying the impact of major claim events (e.g. catastrophe)
- → Disadvantages → requires estimation of future development
  - → Provides a less accurate match of premiums + losses (compared to PY)

→ Policy Year Aggregation
- → Advantages → Provides the best match between losses & premiums
  - → useful for identifying the impact of underwriting or pricing changes
- → Disadvantages → Development on IBNR claims is excluded



→ Report year aggregation
- → Advantage → Provides more stable data than AY aggregation, as the # of claims is fixed at the end of the year
- → Disadvantage → Development on IBNR claims is excluded

→ Inforce = vertical line (use full-term premium)

→ Unearned exposures / premium under CY

→ Written Exposure = Earned Exposure + change in unearned exposure
↓ = ↓ + (EOY unearned exposure − BOY unearned exposure) — End/Beginning of year

→ Written premium = EP + change in unearned premium
↓ = ↓ + (EOY unearned premium − BOY unearned premium)

PY → always starts at zero
⇒ WE = EE + UEE; WP = EP + UEP
- → Endorsements + Premium audits count towards PY data
  - → Endorsements get prorated
  - → Premium audits do not

→ EP + endorsements → e.g. $800 \times \frac{2}{12} + 1200 \times \frac{3}{12}$ — follow timeline of WP

→ Aggregating losses
- → First step is always to determine what claims are relevant based on aggregation method + year (ie filter claims) ⇒ then solve!
  - → Incurred loss = Paid + case reserve
  - → CY IL = CY Paid + CY Δ in case reserve
  - → AY IL = Paid + Ending case reserve (for policies w/ accidents in year X only)
  - → PY IL = Paid + Ending case reserve (for policies written in year X only)

2.1 → Ratemaking principles, considerations + adjustments

→ Ratemaking principles → Rates charged by insurance companies should be developed by following these four principles:
- → Principle 1 → A rate is an estimate of the expected value of future costs
- → Principle 2 → A rate provides for all costs associated w/ the transfer of risk (ie balance at aggregate level)
- → Principle 3 → A rate provides for the costs associated w/ an individual risk transfer (ie balance at individual level)
- → Principle 4 → A rate is reasonable + not excessive, inadequate, or unfairly discriminatory if it is an actuarially sound estimate of the expected value of all future costs associated w/ an individual risk transfer

→ Adjusting for shock losses (steps)
- → 1) Options include → Excluding shock losses entirely
  - → Capping all losses based on the minimum amount of insurance offered, typically referred to as the basic limit, or the limit associated w/ the base rate
  - → Capping losses at a different large loss threshold → can be chosen via judgement or quantiles of loss distribution
- → 2) Include provision
  - → First need to consider two conflicting goals though when determining the threshold
    - → 1) Include as many losses as possible
    - → 2) minimize volatility in the ratemaking analysis

SHOCK / CAT
→ In general: ratio $= \frac{\sum \text{losses adjusting for}}{\sum \text{non [illegible] losses}}$
- → factor = 1 + ratio
- → Year$_i$ adj losses = Non [illegible] losses × factor

e.g.) factor $= \frac{\sum \text{XS Loss}}{\sum \text{Non-XS loss}} + 1$ OR factor $= \frac{\sum \text{CAT losses}}{\sum \text{Non-CAT losses}} + 1$ — includes the portion of the shock losses that are under + XS loss threshold

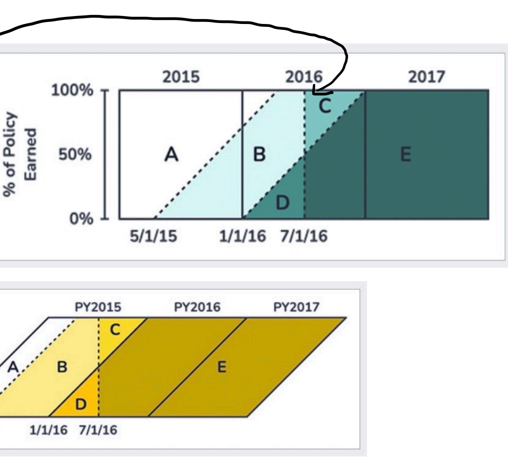
→ TREND BEFORE calculating loading factor ↳ to bring all data to future

→ Reinsurance types → Proportional (ie quota-share, e.g. 70% / 30% ↗ us ↗ them (ie ceded proportion))
- Include in ratemaking analysis? → no / yes (ie reduce losses + premium proportionally OR include net cost of reinsurance as an expense item)
- → Non-proportional → excess-of-loss ⇒ b/c AFFECTS final loss ratio

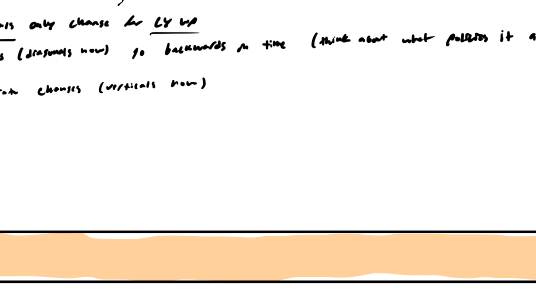
2.2 → Rate + Benefit changes

→ Effects of changes
- → Direct effects are direct + obvious impacts on premium or losses resulting from changes in rates or benefits. Examples include:
  - → ↑ rates ⇒ ↑ premium
  - → ↓ policy limit ⇒ ↓ losses
  - → excluding certain coverages ⇒ ↓ losses
- → Indirect effects are less straightforward

→ CRL is simply using the current rates on all policies
⇒ on-level factor $= \frac{\text{EP @ CRL}}{\text{CY EP}}$ ↳ raw, original EP



→ Policy year



2.3 → Deductibles